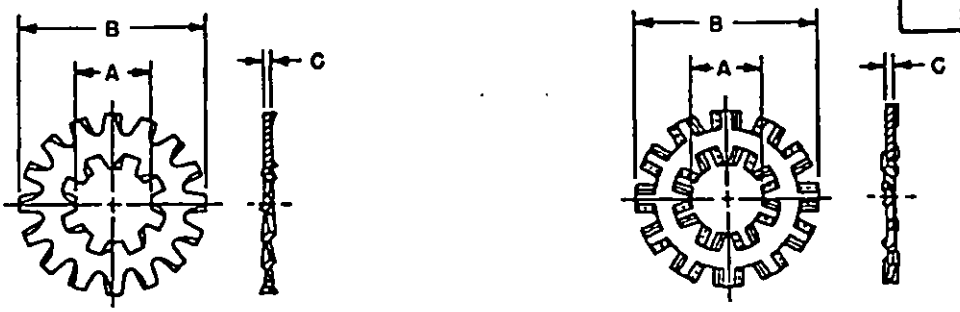
FED. SUP CLASS 5310



OPTIONAL DESIGNS

TABLE I

DASH NO. *	NOM SIZE	A INSIDE DIA MAX	A INSIDE DIA MIN	B OUTSIDE DIA MAX	B OUTSIDE DIA MIN	C THICKNESS MAX	C THICKNESS MIN
51	.112	.123	.115	.475	.460	.021	.016
~~52~~				~~.530~~	~~.495~~	~~.021~~	~~.017~~
~~53~~				~~.610~~	~~.580~~	~~.021~~	~~.017~~
54	.138	.150	.141	.510	.495	.028	.023
~~55~~				~~.610~~	~~.580~~	~~.028~~	~~.023~~
~~56~~				~~.690~~	~~.670~~	~~.028~~	~~.023~~
57	.164	.176	.168	.610	.580	.034	.028
58				.690	.670	.034	.028
59				.760	.740	.034	.028
60	.190	.204	.195	.610	.580	.034	.028
61				.690	.670	.040	.032
62				.760	.740	.040	.032
63				.900	.880	.040	.032
~~64~~	~~.216~~	~~.231~~	~~.221~~	~~.690~~	~~.670~~	~~.040~~	~~.032~~
~~65~~				~~.760~~	~~.725~~	~~.040~~	~~.032~~
~~66~~				~~.900~~	~~.880~~	~~.040~~	~~.032~~
~~67~~				~~.965~~	~~.945~~	~~.045~~	~~.037~~
68	.250	.267	.256	.760	.725	.040	.032
69				.900	.880	.040	.032
70				.985	.965	.045	.037
71				1.070	1.045	.045	.037

DASH NO. *	NOM SIZE	A INSIDE DIA MAX	A INSIDE DIA MIN	B OUTSIDE DIA MAX	B OUTSIDE DIA MIN	C THICKNESS MAX	C THICKNESS MIN
72	.3125	.332	.320	.900	.865	.040	.032
73				.985	.965	.045	.037
74				1.070	1.045	.050	.042
~~75~~				~~1.155~~	~~1.130~~	~~.050~~	~~.042~~
76	.375	.394	.384	.985	.965	.045	.037
77				1.070	1.045	.050	.042
78				1.155	1.130	.050	.042
79				1.260	1.220	.050	.042
~~80~~	~~.4375~~	~~.464~~	~~.448~~	~~1.070~~	~~1.045~~	~~.050~~	~~.042~~
~~81~~				~~1.155~~	~~1.130~~	~~.050~~	~~.042~~
~~82~~				~~1.260~~	~~1.220~~	~~.055~~	~~.047~~
~~83~~				~~1.335~~	~~1.290~~	~~.055~~	~~.047~~
84	.500	.530	.512	1.260	1.220	.055	.047
~~85~~				~~1.335~~	~~1.290~~	~~.055~~	~~.047~~
86				1.410	1.380	.060	.052
87				1.620	1.590	.067	.059
~~88~~	~~.5625~~	~~.596~~	~~.576~~	~~1.335~~	~~1.290~~	~~.055~~	~~.047~~
~~89~~				~~1.430~~	~~1.380~~	~~.060~~	~~.052~~
~~90~~				~~1.620~~	~~1.590~~	~~.067~~	~~.059~~
~~91~~				~~1.830~~	~~1.797~~	~~.067~~	~~.059~~
92	.625	.663	.640	1.410	1.380	.060	.052
93				1.620	1.590	.067	.059
~~94~~				~~1.830~~	~~1.797~~	~~.067~~	~~.059~~
~~95~~				~~1.975~~	~~1.935~~	~~.067~~	~~.059~~

NOTES:

1. MATERIAL: Carbon steel in accordance with Fed. Std. No. 66, AISI 1050 to 1064.
2. PROTECTIVE COATING: Cadmium plated in accordance with QQ-P-416, Type II, Class 3.
3. DIMENSIONS: All dimensions in Table I are in inches. All dimensions in Table II are in millemetres.
4. PART NUMBER: The MS part number consists of the MS number, plus the dash number. Example: MS45904-51.
5. Referenced documents shall be of the issue in effect on date of invitations for bid.
6. For design feature purposes, this standard takes precedence over procurement documents referenced herein.
7. Dash numbers 1 through 45 are canceled after 26 June 1968 and superseded by dash numbers 51 through 95, respectively.
- *8. Lined-out washers are inactive for new design.

(B)

(D) For changes see sheets 1 and 2.

REVIEWER: AV, IS, MU, NS
USER: AS, AT, MC, SH, WY, YD

This military standard is approved for use by all Departments and Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document.

APPROVED 8 MAR 61 REVISED (A) 26 JUN 58 (B) 6 FEB 1974

P.A. MC	TITLE	MILITARY STANDARD
Other Cust 05 82	WASHER, LOCK, INTERNAL AND EXTERNAL TOOTH	MS 45904
PROCUREMENT SPECIFICATION FF-W-100	SUPERSEDES: BEAIS	SHEET 1 OF 2

DD FORM 672-1

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FED. SUP CLASS 5310

(B)

TABLE II
METRIC TRANSLATION OF TABLE I

DASH NO.	NOM SIZE	A INSIDE DIA MAX	A INSIDE DIA MIN	B OUTSIDE DIA MAX	B OUTSIDE DIA MIN	C THICKNESS MAX	C THICKNESS MIN
51	2.84	3.12	2.92	12.06	11.68	.53	.41
~~52~~				~~12.95~~	~~12.57~~	~~.53~~	~~.43~~
~~53~~				~~13.49~~	~~13.03~~	~~.53~~	~~.43~~
54	3.51	3.81	3.58	12.95	12.57	.71	.58
~~55~~				~~15.49~~	~~14.73~~	~~.71~~	~~.58~~
~~56~~				~~17.53~~	~~17.02~~	~~.71~~	~~.58~~
57	4.17	4.47	4.27	15.49	14.73	.86	.71
58				17.53	17.02	.86	.71
59				19.30	18.80	.86	.71
60	4.83	5.18	4.95	15.49	14.73	.86	.71
61				17.53	17.02	1.02	.81
62				19.30	18.80	1.02	.81
63				22.86	22.35	1.02	.81
~~64~~	~~5.49~~	~~5.87~~	~~5.61~~	~~19.53~~	~~19.02~~	~~1.02~~	~~.81~~
~~65~~				~~19.30~~	~~18.42~~	~~1.02~~	~~.81~~
~~66~~				~~22.86~~	~~22.35~~	~~1.02~~	~~.81~~
~~67~~				~~25.02~~	~~24.51~~	~~1.14~~	~~.94~~
68	6.35	6.78	6.50	19.30	18.42	1.02	.81
69				22.86	22.35	1.02	.81
70				25.02	24.51	1.14	.94
71				27.18	26.54	1.14	.94
72	7.94	8.43	8.13	22.86	21.97	1.02	.81
73				25.02	24.51	1.14	.94
74				27.18	26.54	1.27	1.07
~~75~~				~~29.34~~	~~28.70~~	~~1.27~~	~~1.07~~
76	9.52	10.11	9.75	25.02	24.51	1.14	.94
77				27.18	26.54	1.27	1.07
78				29.34	28.70	1.27	1.07
79				32.00	30.99	1.27	1.07
~~80~~	~~11.11~~	~~11.79~~	~~11.38~~	~~27.18~~	~~26.54~~	~~1.27~~	~~1.07~~
~~81~~				~~29.34~~	~~28.70~~	~~1.27~~	~~1.07~~
~~82~~				~~32.00~~	~~30.99~~	~~1.40~~	~~1.19~~
~~83~~				~~33.40~~	~~32.77~~	~~1.40~~	~~1.19~~
84	12.70	13.46	13.00	32.00	30.99	1.40	1.19
~~85~~				~~33.40~~	~~32.77~~	~~1.40~~	~~1.19~~
86				35.81	35.05	1.52	1.32
87				41.15	40.39	1.70	1.50
~~88~~	~~14.29~~	~~15.14~~	~~14.63~~	~~33.40~~	~~32.77~~	~~1.40~~	~~1.19~~
~~89~~				~~36.32~~	~~35.05~~	~~1.52~~	~~1.32~~
~~90~~				~~41.15~~	~~40.39~~	~~1.70~~	~~1.50~~
~~91~~				~~46.48~~	~~45.64~~	~~1.70~~	~~1.50~~
92	15.88	16.84	16.26	35.81	35.05	1.52	1.32
93				41.15	40.39	1.70	1.50
~~94~~				~~46.48~~	~~45.64~~	~~1.70~~	~~1.50~~
~~95~~				~~50.16~~	~~49.15~~	~~1.70~~	~~1.50~~

NOTE:

Lined-out washers are inactive for new design.

REVIEWER: AY, IS, MU, NS
USER: AS, AT, MC, SH, WV, YD

APPROVED 6 MAR 1961 REVISED (A) 26 JUN 1968 (B) FOR CHANGES SEE SHEETS 1 AND 2.

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P.A. MC Other Cust 05 82	TITLE WASHER, LOCK - INTERNAL AND EXTERNAL TOOTH	MILITARY STANDARD MS 45904	
PROCUREMENT SPECIFICATION FF-W-100	SUPERSEDES: BEAX8	SHEET 2 OF 2	

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